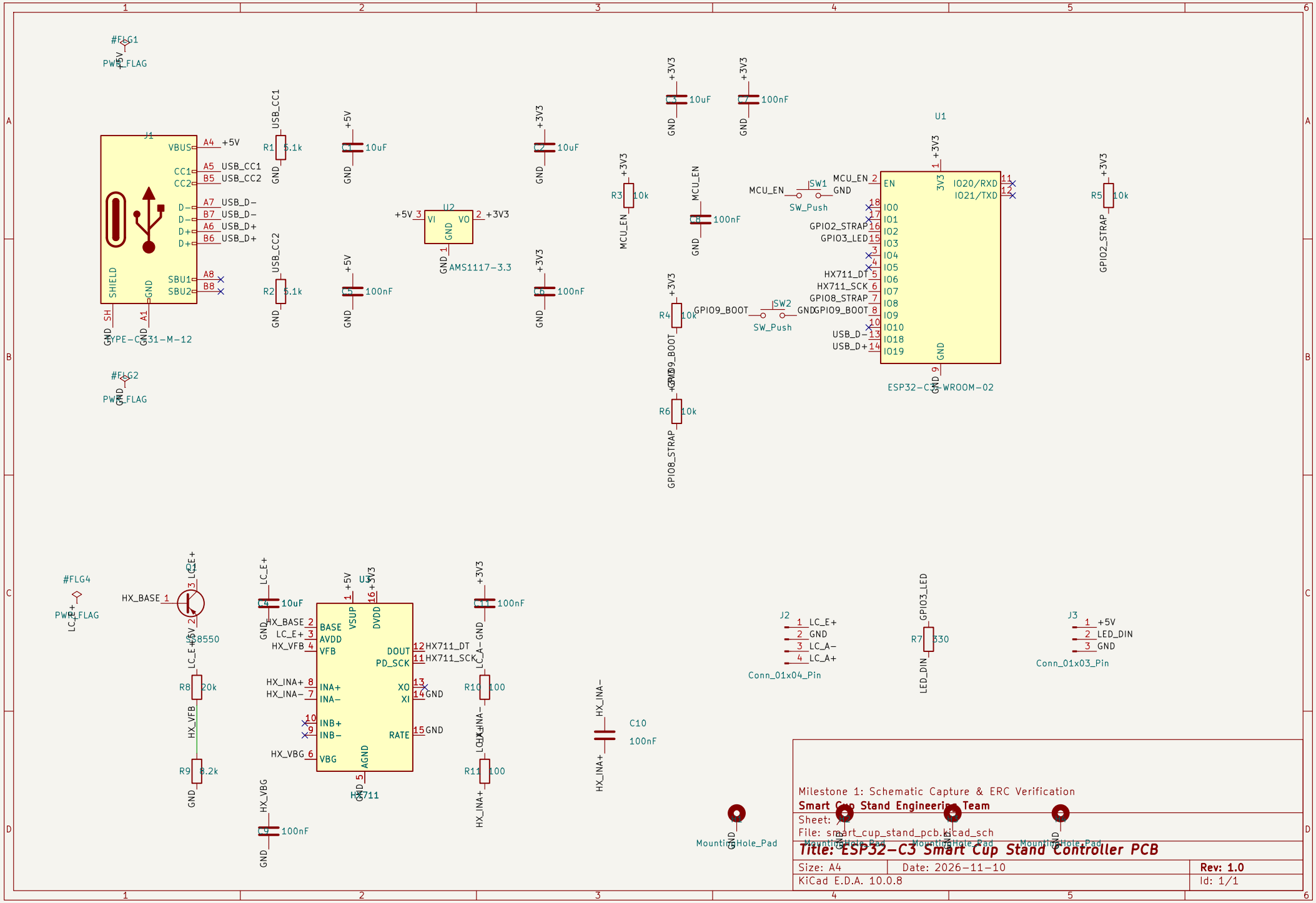

#FLG1
PWR_FLAG
+5V
J1
VBUS
A4 +5V
A5 USB_CC1
B5 USB_CC2
A7 USB_D−
B7 USB_D−
A6 USB_D+
B6 USB_D+
A8
B8
CC1
CC2
D−
D−
D+
D+
SBU1
SBU2
SHIELD
GND
GND_SH
A1
GND
TYPE-C-31-M-12
USB_CC1
R1 5.1k
USB_CC2
R2 5.1k
C1 10uF
C2 10uF
C5 100nF
C6 100nF
U2
+5V 3
VI
VO
2 +3V3
GND 1
GND
AMS1117-3.3
C7 10uF
C8 100nF
C9 100nF
C10 100nF
R3 10k
MCU_EN
SW1
MCU_EN
GND
SW_Push
R4 10k
GPIO9_BOOT
SW2
GPIO9_BOOT
GND
SW_Push
R6 10k
GPIO8_STRAP
U1
3V3
EN
IO0
IO1
GPIO2_STRAP
GPIO3_LED
IO3
IO4
IO5
HX711_DT
HX711_SCK
GPIO8_STRAP
IO7
IO8
IO9
IO10
USB_D−
USB_D+
IO18
IO19
IO20/RXD
IO21/TXD
GND
ESP32-C3-WROOM-02
R5 10k
GPIO2_STRAP
#FLG2
PWR_FLAG
#FLG4
PWR_FLAG
LC_E+
HX_BASE
Q1
S8550
R8 20k
R9 8.2k
HX_VFB
C4 10uF
U3
HX711
BASE
AVDD
VFB
VSUP
DVDD
DOUT
PD_SCK
INA+
INA−
INB+
INB−
VBG
AGND
XO
XI
RATE
HX711_DT
HX711_SCK
C11 100nF
R10 100
R11 100
LC_A−
LC_A+
C9 100nF
HX_VBG
J2
1 LC_E+
2 GND
3 LC_A−
4 LC_A+
Conn_01x04_Pin
R7 330
LED_DIN
GPIO3_LED
J3
1 +5V
2 LED_DIN
3 GND
Conn_01x03_Pin
MountingHole_Pad
MountingHole_Pad
MountingHole_Pad
MountingHole_Pad
Milestone 1: Schematic Capture & ERC Verification
Smart Cup Stand Engineering Team
Sheet:
File: smart_cup_stand_pcb.kicad_sch
Title: ESP32-C3 Smart Cup Stand Controller PCB
Size: A4
Date: 2026-11-10
Rev: 1.0
KiCad E.D.A. 10.0.8
Id: 1/1